

6.3 From Problem to Business Idea

اچھا کاروباری خیال روزمرہ زندگی کے مشاہدے سے شروع ہوتا ہے۔

CS XII

Chapter 6: Entrepreneurship in the Digital Age

Lecture 04 of 16

6.3 From Problem to Business Idea

LEARN IT

A good business idea usually starts from **observing daily life**. Look around your home, college, market, village, or city and ask these simple questions:

1. **What problem do people face?**
2. **Who faces this problem?**
3. **How often does it happen?**
4. **What do people currently do to solve it?**
5. **Can technology help in a simple way?**

A short way to remember them: **What? Who? How often? Current fix? Tech help?**

WORKED EXAMPLE

Example scenario (from the source): study material before exams.

Students cannot find study material before exams. The quick answer is "students are careless", but that is not the real issue.

Apply the five questions:

1. **What problem?** Students cannot find notes before exams.
2. **Who?** Students of a class or college.
3. **How often?** Before every test and exam.
4. **Current fix?** Notes are shared in many places: WhatsApp, email, photocopies, and class groups. Students search all of them.
5. **Tech help?** Yes: a simple **subject-wise notes organizer** where all files are placed in one location (for example, one shared Google Drive folder per subject).

The real problem is **scattered notes**, and the idea solves exactly that.

Scope of the problem

Avoid choosing **very large problems** at the start, such as "improving the whole education system". A better problem is **smaller and clear**, such as helping Grade XII students track **practical file submission dates**. Smaller problems are **easier to test and improve**.

Table 6.2: From problem to idea

The journey from observing a problem to creating an idea follows a structured path.

Step	What Students Do	Example
Observe	Notice repeated problems around home, college, or market.	Students forget practical file dates.
Ask Users	Talk to people who face the problem.	Ask five classmates how they remember deadlines.
Select One Problem	Choose one clear problem only.	Practical file reminder.
Suggest Solution	Think of a simple digital or paper-based solution.	Reminder form or app screen.
Test Early	Show the idea to users before final development.	Ask classmates if they will use it.

After this, the idea moves along the full path that the rest of this chapter follows:



REMEMBER IT

Mnemonic for Table 6.2: "Only Ask Simple Solutions Today": Observe, Ask users, Select one problem, Suggest solution, Test early.

COMMON MISTAKE

Mistake 1: Choosing a huge problem ("fix traffic in Karachi"). **Fix:** Narrow it ("help students of one college know when their van will arrive").

Mistake 2: Skipping "Ask Users" and guessing what people need. **Fix:** Talk to at least five real people who face the problem.

اردو میں سمجھیں

اچھا کاروباری خیال روزمرہ زندگی کے مشاہدے سے شروع ہوتا ہے۔ پوچھیں: مسئلہ کیا ہے، کس کو ہے، کتنی بار ہوتا ہے، لوگ ابھی کیا کرتے ہیں، اور کیا ٹیکنالوجی آسان طریقے سے مدد کر سکتی ہے؟ شروع میں چھوٹا اور واضح مسئلہ چنیں، کیونکہ اسے آزمانا اور بہتر بنانا آسان ہوتا ہے۔

CHECK YOURSELF

1. Write the five steps of Table 6.2 in order.
2. Why is "improving the whole education system" a poor first problem?
3. In the study-material example, what is the real problem?

Answers: 1. Observe, Ask Users, Select One Problem, Suggest Solution, Test Early. 2. It is too large and unclear; small, clear problems are easier to test and improve. 3. Notes are scattered in many places (WhatsApp, email, photocopies, class groups), not that students are careless.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An IDE is:

- a. One app to write, run and debug code
- b. A printer
- c. A sort
- d. A cable

MCQ 2. VS Code is:

- a. A popular editor/IDE
- b. A DBMS
- c. An OSI layer
- d. A gate

MCQ 3. The terminal/run button is for:

- a. Executing the program
- b. Drawing pie charts only
- c. Sending TCP SYN only
- d. K-maps

MCQ 4. Syntax highlighting helps you:

- a. See keywords and mistakes
- b. Sort faster
- c. Encrypt disks
- d. Route packets

MCQ 5. A file explorer in VS Code shows:

- a. Project files
- b. OSI layer 1
- c. NAND groups
- d. Beachhead users

MCQ 6. Debugging in an IDE uses:

- a. Breakpoints and variable watch
- b. Only bubble sort
- c. Only pie charts
- d. Only Agile stickers

Extra short questions

1. Define IDE.
2. Name four VS Code areas.
3. Why not use Notepad only?
4. What is an extension?

Extra long question

1. Describe how you would create, run and debug a first Python file in VS Code. Include two interface elements from the notes.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Integrated Development Environment: editor + run + debug + files in one place.
2. Activity bar/sidebar (files), editor, terminal, status bar (and extensions).
3. No easy run, no highlighting, no debugger.
4. An add-on (Python plugin) that teaches VS Code a language.

Long

1. New file .py, type print, Run Python, read terminal. Mention top menu, explorer, terminal panel. Breakpoint on a wrong variable.